

CHAPTER-2: INTRODUCTION OF PROJECT

2.1 Introduction of problem

2.1.1 Problem statement

- The tenure can range from a few months to a few years. The main disadvantage of an EMI is that it can be quite expensive.

2.1.2 Scope of Proposed System

- The main benefit of an EMI calculator for borrowers is that they get to know how much money they will need to pay towards their loan each month so that they do not end up straining themselves financially and end up in financial distress.
- Loan EMI calculator helps the borrower decide on an EMI amount that can be comfortably repaid over the tenure of the loan.
- User can calculate EMI for home loan, car loan, personal loan, education loan or any other fully amortizing loan using this calculator.

2.2 Environment Description

2.2.1 Hardware and Software Requirement:

- **Software requirement:**

- Software requirement to develop system:
Operating System: Windows 10 or higher version
Developing Language: Android , JAVA
Graphics software: Canva , Adobe Photoshop
- Software requirement to use system:
Operating System: Android 8 or Higher

- **Hardware requirement:**

- Hardware requirement to develop system:
Processor: Intel core i5 or Higher
RAM: 8 GB
Storage: 25 GB
- Hardware requirement to use
system:RAM (Minimum): 1 GB
Storage (Minimum): 200 MB

2.2.2 Technology Used

Following are the Technologies used to develop “EMI Calculator”.

JAVA :



James Gosling, often called as the father of Java pioneered Java in June 1991 as a project called ‘Oak.’ Gosling aimed to develop a virtual machine and language with a well-known notation like C but with more precision and simplicity than C/C++. In 1995, Java 1.0 was the first public execution. It pledged ‘Write Once, Run Anywhere’ on popular platforms with free runtimes. It was very safe and configurable, with security restricting network and file access. In a stable ‘applet’ setup, the significant web browsers soon implemented it in their standard settings.

In 1997, Sun reached out to the ISO/IEC JTC1 and then Ecma International to formalize Java, but they quickly withdrew. Java continues to be a de facto proprietary standard regulated by the Java Community Process. With the revenue generated by a new vision, such as the Java Enterprise Framework, Sun made several Java implementations free of charge. The critical difference is that the compiler is not present in the JRE, differentiating between its Software Development Kit (SDK) and JRE (JRE).

On 13 November 2006, Sun launched a considerable amount of Java in the GNU General Public License as free and open-source software (GPL) (GPL).

ANDROID :



Android, operating system for cellular telephones and tablet computers. Android began in 2003 as a project of the American technology company Android Inc., to develop an operating system for digital cameras. In 2004 the project changed to become an operating system for smartphones. Android Inc., was bought by the American search engine company Google inc., in 2005. At Google, the Android team decided to base their project on Linux, an open sources operating system for personal computers.

On November 5, 2007, Google announced the founding of the Open Handset Alliance, a consortium of dozens of technology and mobile telephone companies, including Intel Corporation, Motorola, Inc., NVIDIA Corporation, Texas Instruments Incorporated, LG Electronics, Inc., Samsung Electronics, Sprint Nextel Corporation, and T-Mobile (Deutsche Telekom). The consortium was created in order to develop and promote Android as a free open source operating system with support for third-party applications. Android-based devices use wireless networks in order to take full advantage of features such as one-touch Google searches, Google Docs (e.g., word editors, spreadsheets), and Google Earth (satellite mapping software).

CHAPTER-3: THE WHOLE INDUSTRIAL PROCESS AND PROBLEM STUDY

3.1 System Planning

3.1.1 Expected Modules

- **User:**
 - **Loan Amount:**
 - ✓ The principal amount that you wish to borrow should be entered into the EMI calculator in the space which says loan amount.
 - **Interest:**
 - ✓ It is the rate at which the interest is charged on the amount lent to the borrower. The interest rate also varies from one type of loan to another, on the basis of the amount lent and from one lender to another
 - **Year / Month:**
 - ✓ The loan tenure you choose is inversely proportional to the EMI. The longer the tenure, the lower the EMI. However, with longer tenure you may end up paying more as interest. The loan tenure options usually range between 1
 - **Rate Us:**
 - ✓ In this the we allow our user to rate us accordingly as there satisfaction
 - **Share App:**
 - ✓ User are free to share our app with others as its completely user friendly.

3.1.2 Limitation of project

- It is only used to calculate vast amount of data.
- The risk of default and loss of collateral.

3.2 Use Case Diagram

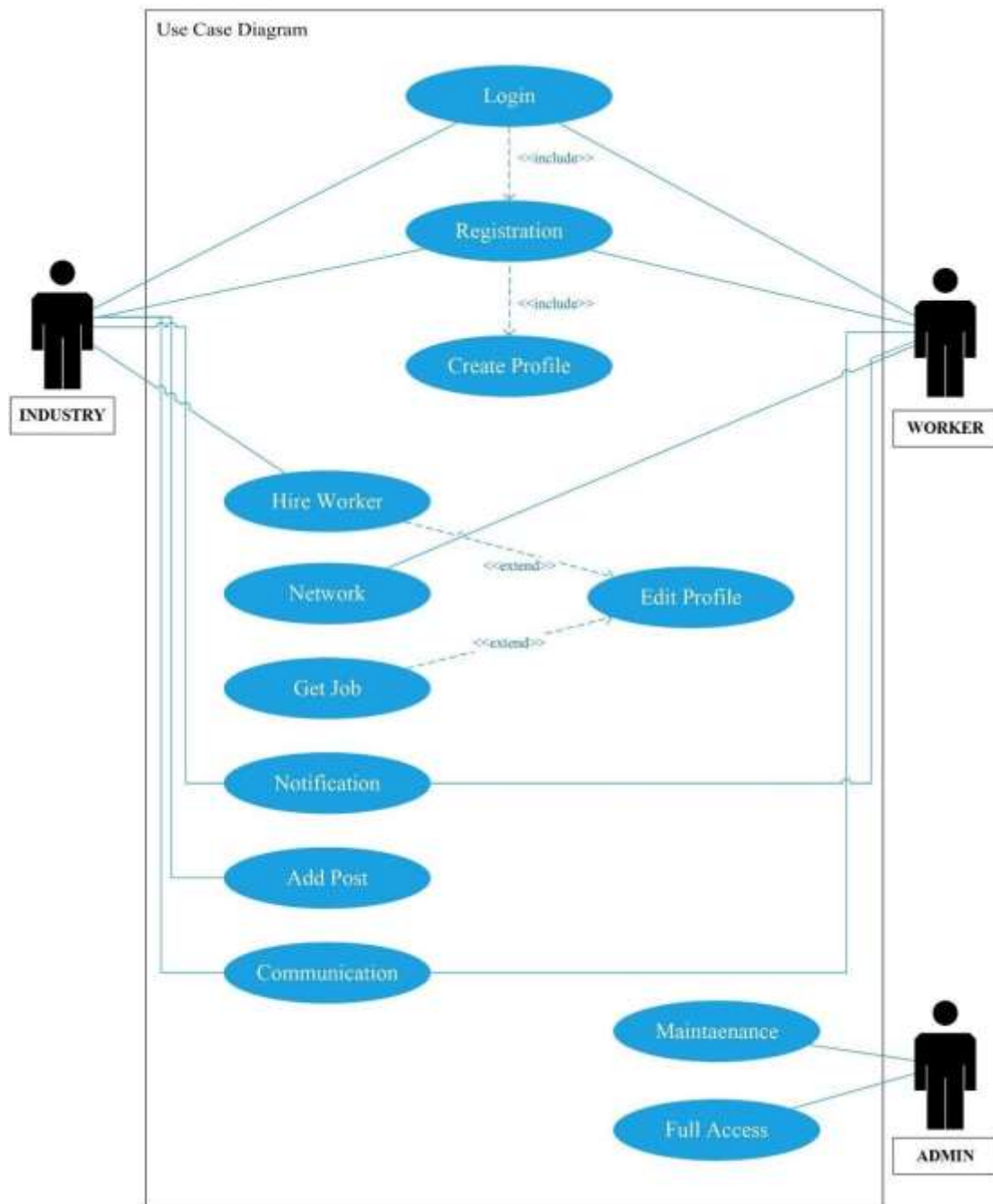


Figure 3.2.1

3.3 Activity Diagram

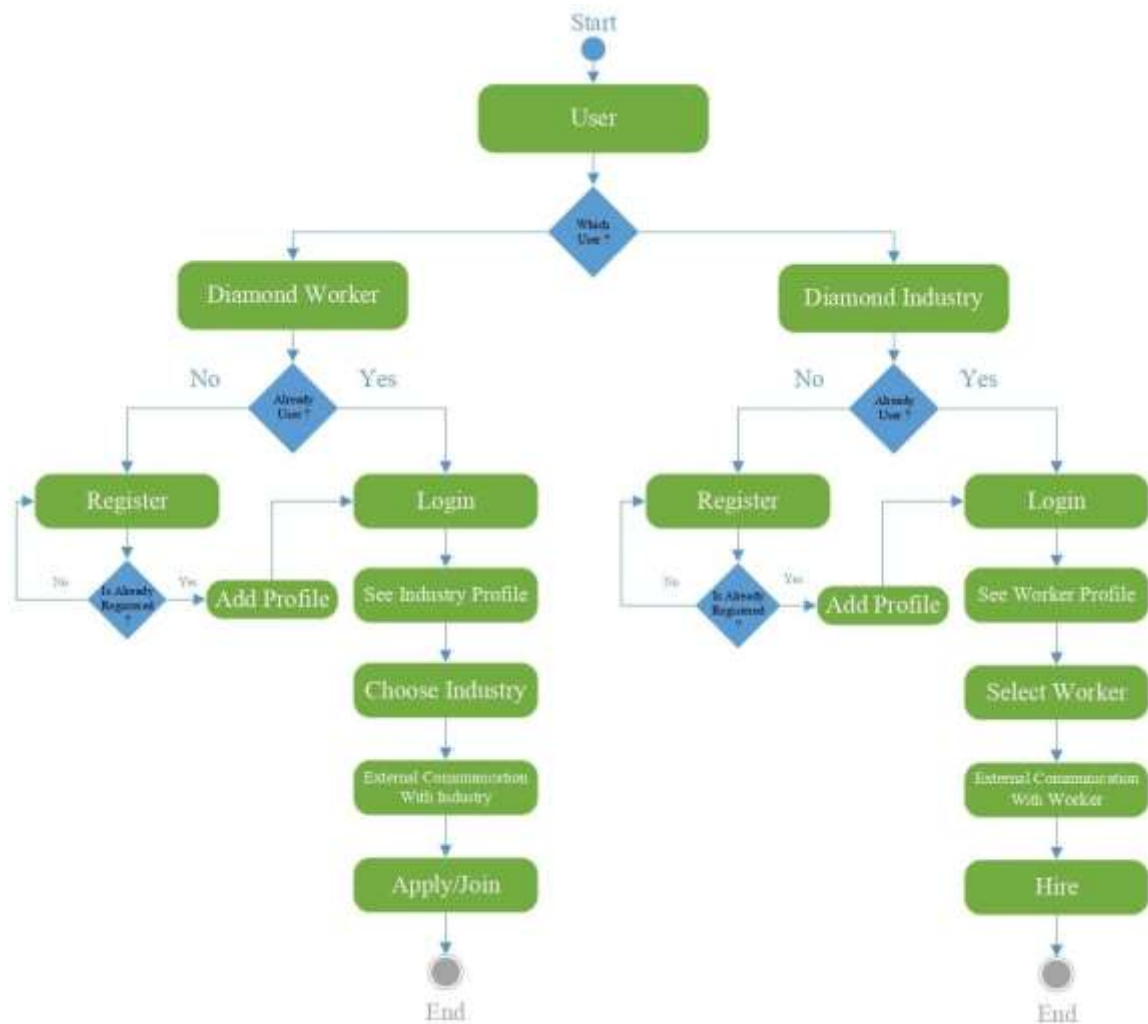


Figure 3.3.1

3.4 Data Flow Diagram

- DFD Level 0

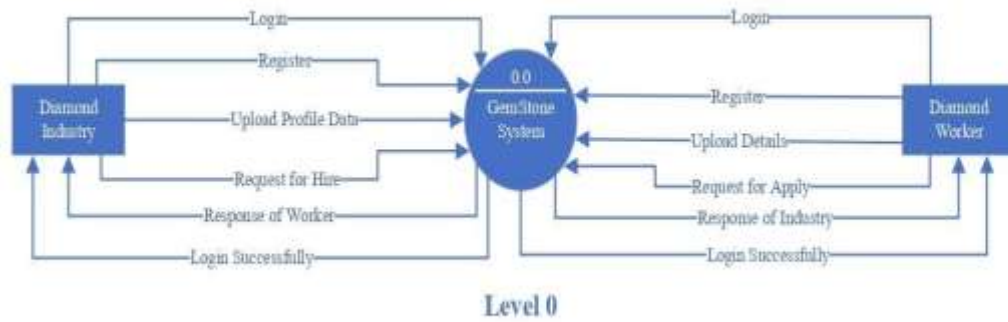


Figure 3.4.1

- DFD Level 1

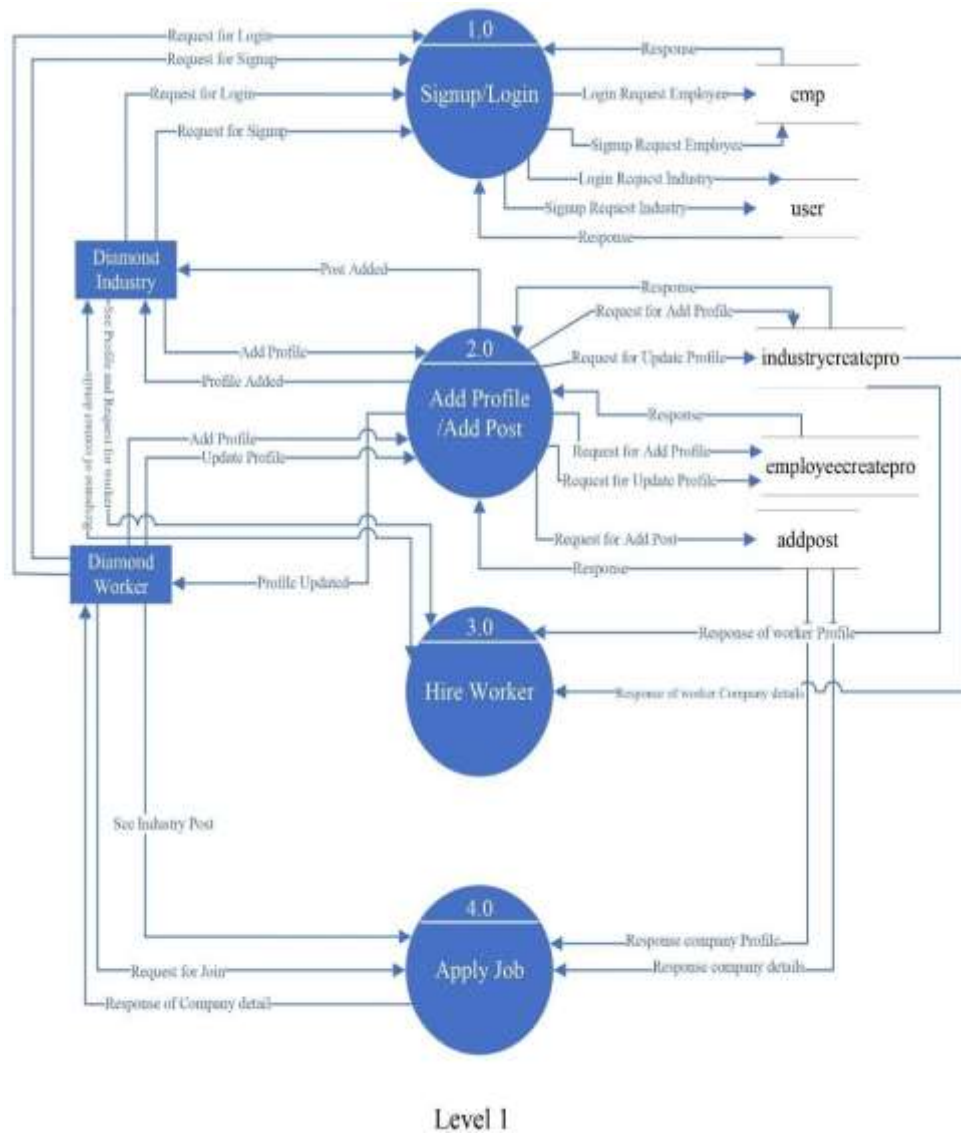


Figure 3.4.2

- DFD Level 2

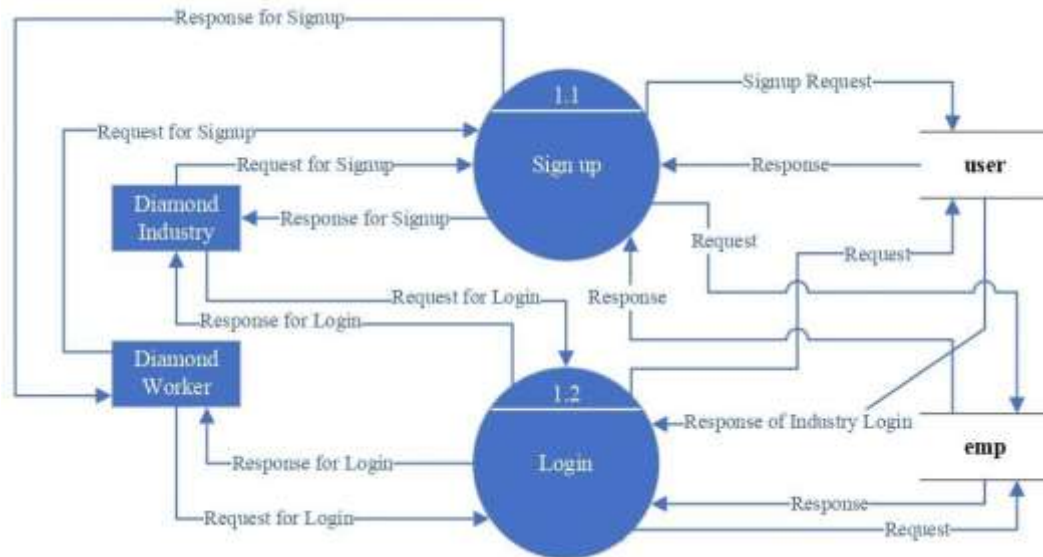


Figure 3.4.3

CHAPTER-4: THE PROBLEM SOLUTION OUTLINE

1. Input Design/Output Design (Industry)



OnLoad



Homepage

The Startapp screen shows the "EMI Calculator" title and a "RESET" button. It contains input fields for "Loan Amount" (100000), "Interest" (3.5), "Tenure" (8), and "Age" (6). A red "CALCULATE" button is positioned below the inputs. The results section displays "Monthly EMI: 1740.09", "Total Interest: 77489.18", and "Total Payment: 177489.18". At the bottom, there are "HISTORY" and "DETAILS" buttons.

Field	Value
Loan Amount	100000
Interest	3.5
Tenure	8
Age	6
Monthly EMI	1740.09
Total Interest	77489.18
Total Payment	177489.18

Startapp

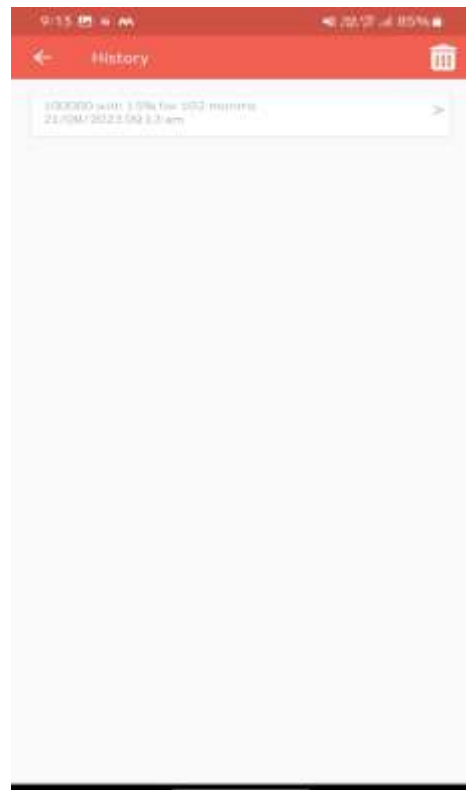
The Detail screen shows a summary of the loan details and a table of monthly payments. The summary includes "Loan Amount: 100000.00", "Interest Rate: 15.00", "Months: 102", "Monthly EMI: 1740.09", "Total Interest: 77489.18", and "Total Payment: 177489.18". Below this is a table with columns for Month, Principle, Interest, and Balance.

Month	Principle	Interest	Balance
1	880	1237	99813
2	887	1242	99012
3	893	1237	98144
4	899	1231	97293
5	915	1235	96402
6	923	1219	95481
7	930	1212	94535

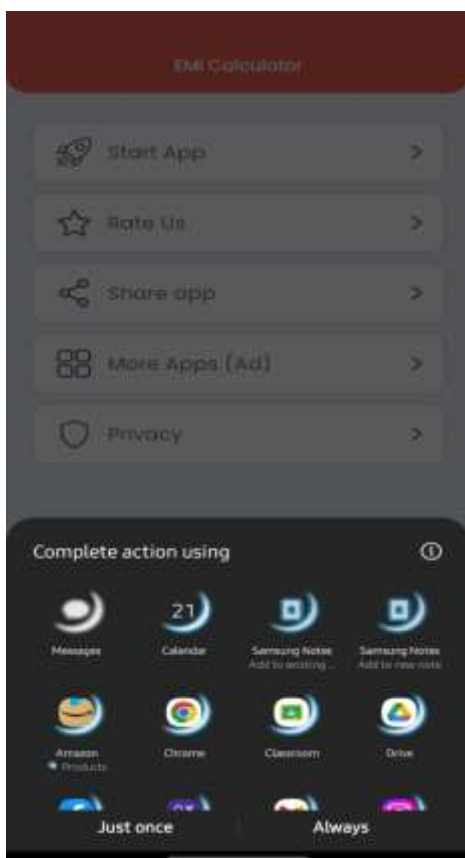
Detail



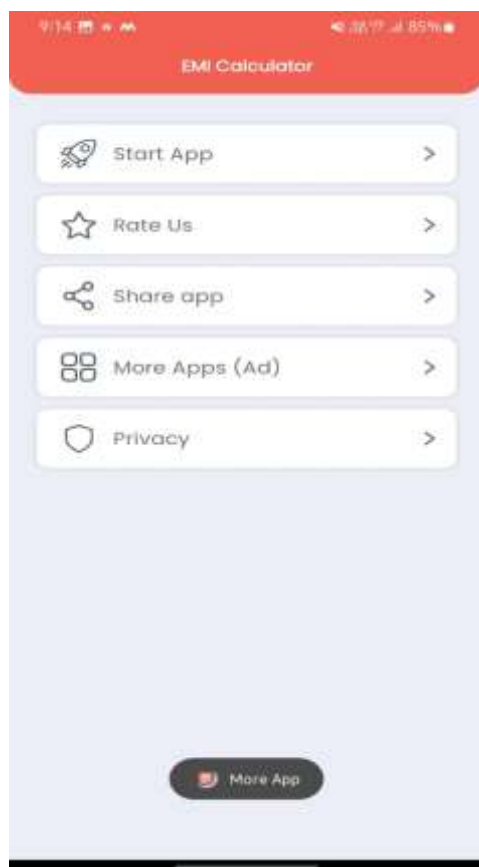
Chart



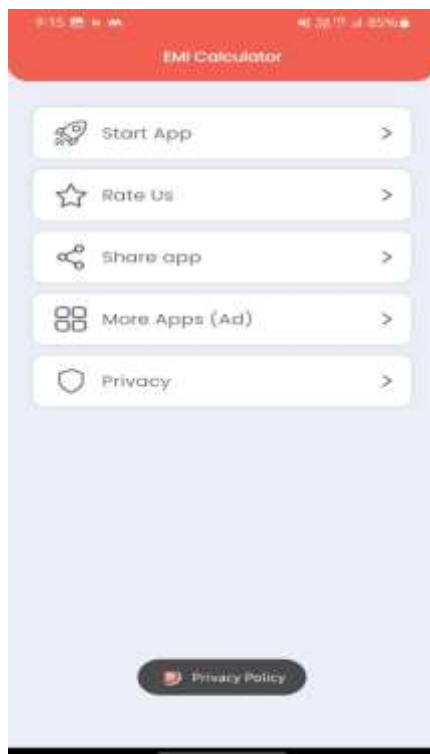
History



Share app



More apps



Privacy

CHAPTER-5: CONCLUSION

- The EMI calculator app provides a convenient way to calculate your monthly loan installments. With its user-friendly interface and accurate calculations it helps you plan your finances effectively. Whether you're applying for a home loan personal loan, or any other type of loan the EMI calculator simplifies the process and provides you with valuable insights.
- Download the app now and take control of your Android app offers a valuable financial tool for borrowers, aiding them in making informed decisions about their loans. the EMI Calculator application developed in Android Studio using Java is an indispensable financial tool for borrowers, facilitating informed decision-making regarding loans. With its user-friendly interface, extensive feature set, and widespread accessibility, it emerges as a valuable addition to the Android app ecosystem, promoting financial literacy and responsible borrowing.

References/Bibliography

Website:

- <https://www.nobroker.in/forum/how-to-calculate-interest-rate-from-emi/>
- <https://www.paisabazaar.com/emi-calculator/>
- <https://www.bajajfinserv.in/interest-calculator>
- <https://emicalculator.net/loan-calculator/>